

Sajjad Hossain Talukder

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PROFILE

Full-stack software engineer building cloud-native backends, AI/LLM-driven automation, and cross-platform applications end to end. ICPC Asia-West finalist and published SDN/NDN networking researcher.

WORK EXPERIENCE

Software Engineer

Nov 2025 – Present

Pivotly, Minnesota, USA (Remote)

- Build AI-powered backend services and agentic LLM workflows that automate high-volume business processes, develop the accompanying web and mobile frontends over REST APIs, and deploy them to production on cloud infrastructure.

Software Engineer

Sep 2024 – Oct 2025

Silicon Orchard Ltd., Dhaka, Bangladesh

- Built and maintained cross-platform (React Native) and native iOS (Swift) applications for global clients, integrating backend APIs and optimizing performance for reliability on unstable networks.

Software Engineer (iOS)

Mar 2024 – Sep 2024

Xotech, Chattogram, Bangladesh

- Developed AI-based iOS applications (Swift, UIKit) backed by PHP/Laravel REST APIs, managing the full lifecycle through to App Store release.

PROJECTS

Grade Now — AI Grading Platform for Students

[Link]

Agentic AI, LLM, Node.js, Next.js, PostgreSQL, AWS

- Solely built and deployed an AI platform that grades student answer scripts against AI-generated rubrics through a parallel, multi-stage agentic pipeline, cutting evaluation time and improving consistency.

Connect My Advocate — Legal Aid Platform

[Link]

Next.js, Node.js, PostgreSQL, Agora SDK, AWS

- Sole engineer behind the complete system — client web, mobile, and admin apps, plus the AWS infrastructure and PostgreSQL backend — for a legal-aid platform with real-time video consultation.

IELTS Pro BD — Mock IELTS Test Platform

[Link]

Next.js, Node.js, PostgreSQL, SSLCommerz

- Architected, shipped, and maintain a live computer-delivered mock-IELTS platform with student management, a flexible test-engine editor, and subscription billing.

SmartWeight — Edge Vision Weighing System

[Link]

Raspberry Pi, Python, OpenCV

- Integrated real-time OpenCV object detection with weight sensors on Raspberry Pi to automate billing and product identification, fully on-device.

SKILLS

- Research & Simulation:** ndnSIM, ns-3, Python/ZeroMQ control plane, D3QN + Prioritized Experience Replay, Federated Learning, Named Data Networking, SDN
- Languages:** JavaScript/TypeScript, Python, Swift, Go, C++, PHP, SQL
- Cloud & DevOps:** AWS, Azure, Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, Firebase, Linux
- Frameworks & Backend:** Next.js, React, Node.js, React Native, SwiftUI/UIKit, Laravel, RESTful APIs
- AI & Systems:** Agentic AI & LLM Integration, AI/ML, OpenCV, Deep Reinforcement Learning (D3QN, PER)

EDUCATION

Bachelor of Science in Computer Science & Engineering

Jan 2019 – Jan 2026

Premier University, Chattogram, Bangladesh

CGPA: 3.98 / 4.00 (1st rank in batch)

PUBLICATIONS

PDAF: Performance-Driven Adaptive Forwarding in SDN-Assisted NDN-MANETs [Link]
S. H. Talukder, S. M. A. Iqbal, E. Barua
IEEE Transactions on Mobile Computing, 2026, in press, doi: 10.1109/TMC.2026.3722116

Empowering Bengali Language in Drone Control with Artificial Neural Networks [Link]
S. H. Talukder, N. Rezoana, T. Mahmud, N. Basnin, S. Chowdhury, M. S. Hossain, K. Andersson
Book chapter in Applied Machine Learning on Sensing Technologies, CRC Press / Taylor & Francis, pp. 81-93, 2026, doi: 10.1201/9781003506218-6

AWARDS & ACHIEVEMENTS

- **Competitive Programming — ICPC:** ICPC Asia West Championship Finalist, 2022 (Team PUC Eternals), ICPC Dhaka Regional — Main Round 52nd (2022), 78th (2023), 93rd (2021); Preliminary 76th of 2481 (2023)
- **Competitive Programming — Ratings:** Codeforces: Specialist (max rating 1417), 1000+ problems solved, CodeChef: 3-star (max rating 1721); also active on LeetCode, LightOJ and TOPH
- **Scholarships:** Merit-Based Scholarship, Premier University

CERTIFICATIONS & TEST SCORES

IELTS Academic — Overall Band 7.0 (CEFR C1) [Link]
British Council / IDP · Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0, 2026

ACTIVITIES & LEADERSHIP

Manager and Trainer (Competitive Programming)	Jan 2021 – Dec 2023
<i>Premier University Computer Club</i>	
Ran data-structures and algorithms bootcamps for competitive-programming teams.	
Program Secretary	Jan 2022 – Dec 2022
<i>Premier University Robotics Club</i>	
Ran programming classes for club members and arranged and managed the club's events.	